Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.7 – Design Challenge — Light the Display

Unit 2: Electricity & Energy

Learning intentions

The brief: the school's winter display needs **3 identical LEDs**. Each LED needs exactly **2 V** and **0.3 A** to glow safely — too much of either and it burns out. The only power supply available is a **6 V** battery pack. **Design the circuit.**

No new ideas today — everything you need is from the last six lessons. This is where you put it all together, on your own.

Step 1 — Choose your arrangement

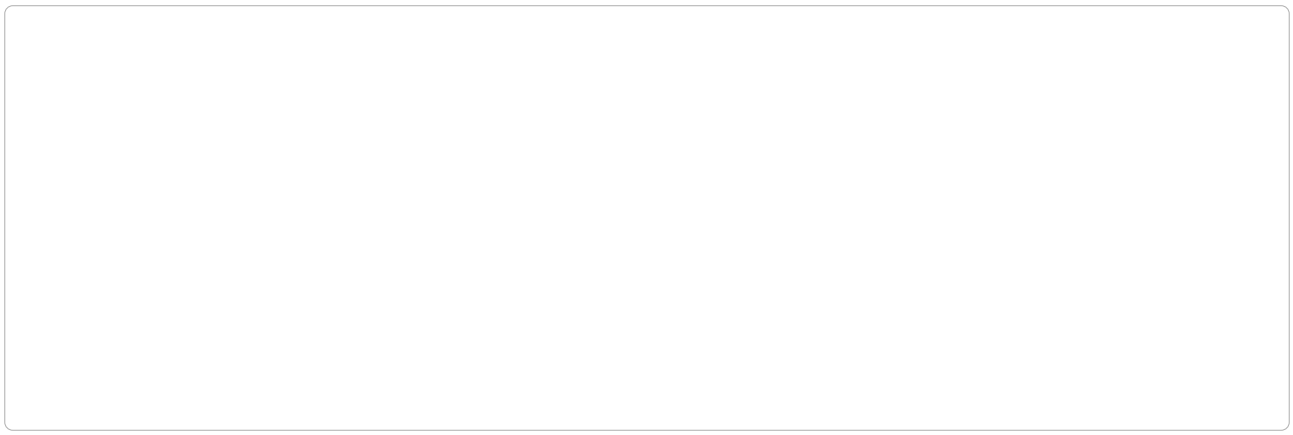
1) **Decide** series or parallel for your 3 LEDs, and justify your choice using **one** fact from Lesson 3 or 4 about what happens if **one** LED fails.

Step 2 — Check the voltage

2) **Calculate** whether your chosen arrangement delivers exactly 2 V to each LED straight from the 6 V supply, or whether a resistor is needed to drop the spare voltage. If a resistor is needed, calculate its value, using 0.3 A as the current through it.

Step 3 — Draw it

3) **Draw** the full circuit diagram for your design: the battery, a switch, all three LEDs, and any resistors your Step 2 answer needs. Label every symbol.



Name: _____ **Date:** _____

Step 4 — Check it

4) **Draw** an ammeter and a voltmeter onto your Step 3 diagram so that, together, they **prove** one LED is getting exactly 2 V and 0.3 A — not just that the whole display lights up.

Step 5 — Power

5) **Calculate** the **total** power the battery supplies to your design.

Extension — which design actually wastes energy

Of the power you calculated in Step 5, only the power actually used *by the LEDs* ($3 \times 2\,\text{V} \times 0.3\,\text{A} = 1.8\,\text{W}$) does anything useful — the rest is wasted as heat in any resistors.

6) **State** which design — series or parallel — wastes energy as heat, and how much of its total power (as a fraction or percentage) is wasted.

HOMEWORK

The display is being expanded from 3 LEDs to **60**. Would you keep your Step 1 design, or switch to the other arrangement? Write two or three sentences, weighing up what you found in Steps 1–5 at this much larger scale.